2. State whether the following statements are True or False.
   (i) A force is always required to change the speed of motion of an object.
   (ii) Due to friction, the speed of the ball rolling on a flat ground increases.
   (iii) There is no force between two charged objects placed at a small distance apart.
3. Two balloons rubbed with a woollen cloth are brought near each other. What would happen and why?
4. When you drop a coin in a glass of water, it sinks, but when you place a bigger wooden block in water, it floats. Explain.
5. If a ball is thrown upwards, it slows down, stops momentarily, and then falls back to the ground. Name the forces acting on the ball and specify their directions.
   (i) During its upward motion
   (ii) During its downward motion
   (iii) At its topmost position
6. A ball is released from the point P and moves along an inclined plane and then along a horizontal surface as shown in the Fig. 5.16. It comes to stop at the point A on the horizontal surface. Think of a way so that when the ball is released from the same point P, it stops (i) before the point A (ii) after crossing the point A.

Fig. 5.16

7. Why do we sometimes slip on smooth surfaces like ice or polished floors? Explain.
8. Is any force being applied to an object in a non-uniform motion?
9. The weight of an object on the Moon becomes one-sixth of its weight on the Earth. What causes this change? Does the mass of the object also become one-sixth of its mass on the Earth?

Reflect on the questions framed by your friends and try to answer ...

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10. Three objects 1, 2, and 3 of the same size and shape but made of different materials are placed in the water. They dip to different depths as shown in Fig. 5.17. If the weights of the three objects 1, 2, and 3 are w_1, w_2, and w_3, respectively, then
    (i) $w_1 = w_2 = w_3$
    (ii) $w_1 > w_2 > w_3$
    (iii) $w_2 > w_3 > w_1$
    (iv) $w_3 > w_1 > w_2$

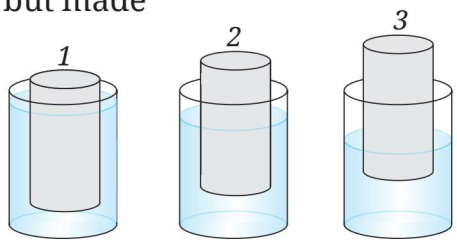
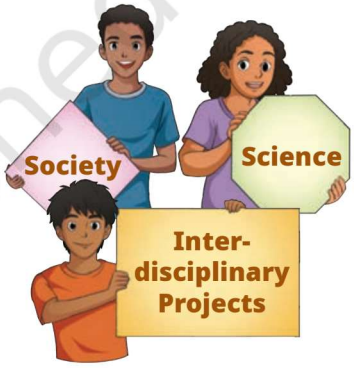


Fig. 5.17

Discover, design, and debate



- Collect objects made of different materials, such as plastic, wool, silk, rubber, polythene sheet, paper, and metals. Rub one material with another and check if it attracts small pieces of paper or not, that is, whether it gets charged or not. Record your observations in a systematic manner and write a research paper.
- Imagine a scenario where the gravity disappears. Develop a story. Create a cartoon strip to present your story.
- Organise a discussion in your class on the topic: Friction—a necessity or a problem? Make a note of the discussion and state where friction is a necessity and when it is a problem.
- Make your own spring balance with the help of your teacher and calibrate it using standard weights. Now measure the weights of different objects and calculate the ratio of the weight and mass of different objects. Do you observe a pattern?
- An electroscope is a device which can determine whether an object is electrically charged. You can make your own electroscope (Fig. 5.18) in your class with the help of your teacher, test the device. Explore in what other ways you may use this electroscope.

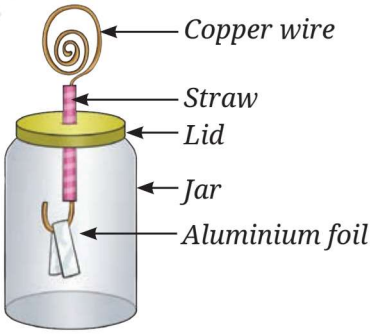


Fig. 5.18

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